The Impact of Online Scandals on Brand Perception and Market Value: The Case of Amazon's Tax Avoidance Controversy

Ali Al-Rikabi (10166070)

Daniela Mañozca Cruz (30262558)

David Fakolujo (30273636)

Luisa Alejandra Sierra Guerra (30261956)

University Of Calgary

Faculty Of Science

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Introduction

Corporate reputation is a critical intangible asset that underpins stakeholder trust, particularly among consumers. In today's hyperconnected digital ecosystem, a brand's online presence plays a pivotal role in shaping public opinion and influencing brand perception, which refers to how consumers view and emotionally respond to a company's identity and behavior. Online scandals, amplified through social media due to its immediacy and global reach, can significantly disrupt consumer sentiment, often leading to rapid and widespread reputational damage.

These scandals occur when brands face negative attention online, sparking emotions like distrust, frustration, or outrage. Such reactions can lead to harmful consumer actions, like negative word-of-mouth, brand avoidance, switching to competitors, or boycotts. The root cause is often a disconnect between societal expectations and a company's actions, such as mistreating employees or exploiting loopholes (e.g., tax avoidance). Ultimately, reputational damage can erode brand equity, stakeholder trust, and financial performance.

This project focuses on public policy-related brand crises, using Amazon's 2018 tax controversy as a case study. In that year, Amazon was widely criticized for paying minimal federal taxes in the United States, despite substantial earnings. The case is emblematic due to its intersection of fiscal policy, corporate ethics, and socio-political sensitivity, especially given the universal obligation of taxpayers and the emotional reactions such inequities provoke.

To capture the multidimensional impact of the scandal, this study will perform a sentiment analysis using a customized framework with ten categories tailored to reputational

crisis. It will also conduct a comparative analysis of Amazon's market value before, during, and after the event, using an external benchmark to isolate its effects. Additionally, machine learning techniques will be applied for clustering consumer perceptions and reducing data dimensionality to extract clearer insights.

Through this integrative approach, this project aims to "ANALYZE THE IMPACT OF ONLINE SCANDALS ON BRAND PERCEPTION AND MARKET VALUE" in the case of Amazon's 2018 tax controversy, providing data-driven insights for marketing professionals, policy analysts, and corporate strategists managing reputational risk in the digital era.

Analysis Questions

Main Objective: “ANALYZE THE IMPACT OF ONLINE SCANDALS ON BRAND PERCEPTION AND MARKET VALUE” in the case of Amazon’s 2018 tax controversy.

EDA Objectives and Questions

1. **Obj**: Identify temporal trends in online searches related to Amazon and taxes, as well as Amazon’s stock prices.

   1.1 Q: What were the search patterns on Google related to Amazon and taxes? Understanding search trends over time reveals shifts in public attention and engagement with the scandal, which helps contextualize changes in consumer awareness and concern.

   1.2 Q: How did Amazon’s stock price evolve before, during, and after the scandal? Examining stock price dynamics across these periods provides insight into how financial markets responded to reputational risk and public scrutiny.

Statistical Data Objectives and Questions

2. **Obj**: Explore the correlation between public online interest in Amazon and its commercial value.

   2.1 Q: Is there a correlation between online interest and Amazon’s commercial value? Online search interest is a proxy for consumer and investor attention. Correlating this interest with market valuation can reveal the economic consequences of reputational events.

Machine Learning Objectives and Questions

3. **Obj**: Assess how search engine trends influence Amazon’s commercial value.

   3.1 Q: Can trends in search engines impact Amazon’s stock price?

Using machine learning to detect patterns and clusters in search trends and stock data can help identify complex, non-linear relationships that traditional methods might overlook.

Sentiment Analysis Objectives and Questions

Comments on a video explicitly focused on Amazon serve as direct or indirect indicators of public perception. Sentiment polarity (positive, negative, neutral) in these comments reflects emotional responses and can be linked to shifts in brand reputation.

Based on that objective:

4. **Obj**: Analyze the general tone, sentiment polarity, and thematic focus of viewer comments related to Amazon's tax scandal in order to assess public opinion trends.

   4.1 Q: How do the sentiment polarity and thematic patterns in viewer comments reflect public opinion regarding Amazon's tax scandal?

   - Identify the primary topics associated with Amazon's tax scandal.(Word frequency plot)
   - Evaluate the overall sentiment polarity and subjectivity of the dataset to assess the general tone of public opinion. (Scatter plot of polarity vs. subjectivity, Bar chart of polarity distribution, Subjectivity vs. objectivity distribution)
   - Explore which categories of viewer comments generated the highest engagement, based on the volume of responses per topic. (Count of comments per topic)

5. **Obj**: Investigate how different categories of comments and their thematic interconnections influence consumer attitudes and shape Amazon's brand perception.

   5.1 Q: To what extent do specific categories of comments and their interconnections influence consumer attitudes and perceptions toward Amazon's brand?

- Determine how specific categories of comments contribute to shaping consumer attitudes toward Amazon (Scatter plots for each classified category showing polarity distribution, Topic polarity distribution)
- Map the structure and interconnections among key discussion topics to uncover the dynamics behind audience sentiment. (Centrality and betweenness graph, Correlation matrix or network plot of topics)

Data Sourcing and Justification

The datasets used for this project were obtained from Google Trends https://trends.google.com/trends/, Google Finance https://www.google.com/finance/, and YouTube https://www.youtube.com/watch?v=4f1vNr9j5G8&t=57s. The data was filtered to cover the period from 2015 to 2020, allowing us to capture patterns both before and after Amazon's 2018 tax scandal. Weekly granularity was selected, as it is the minimum temporal resolution provided by Google Trends, enabling us to trace changes in public interest and market activity over time.

For the sentiment analysis component, a publicly available YouTube video discussing Amazon's tax controversy was selected. User comments were extracted using the Instant Data Scraper tool. To ensure ethical standards and compliance with data privacy regulations, personally identifiable information such as usernames and timestamps were removed prior to analysis.

Tracking real-time public responses to corporate scandals can be effectively done by analyzing digital behavior, particularly search engine and social media activity. Spikes in search queries related to a brand and its controversies serve as proxies for public attention and can reflect shifts in brand perception. In parallel, financial indicators such as stock prices provide tangible evidence of market sentiment and investor confidence. Monitoring these indicators before, during, and after a reputational crisis offers a holistic view of the scandal's impact on both consumer perception and firm value.

The dataset compiled from Google Trends and Google Finance consists of 263 weekly observations across nine variables. These are detailed in **Table 1** below.

Table 1.

Description of Variables in the Dataset

Variable Name	Description
Week	Start date of the weekly observation (Sunday of each week)
amazon: (Google Shopping)	Weekly search interest for "Amazon" in Google Shopping (scale 0–100)
amazon: (News search)	Weekly search interest for "Amazon" in Google News (scale 0–100)
tax: (News search)	Weekly search interest for "Tax" in Google News (scale 0–100)
amazon: (Youtube search)	Weekly search interest for "Amazon" on YouTube (scale 0–100)
Open	Amazon stock opening price at the start of the week (in USD)
High	Highest Amazon stock price during the week (in USD)
Low	Lowest Amazon stock price during the week (in USD)
Close	Amazon stock closing price at the end of the week (in USD)
Adj Close	Adjusted closing price (corrected for splits and dividends) (in USD)
Volume	Total number of Amazon shares traded during the week

The YouTube dataset comprises 1,500 user comments related to the selected video. After preprocessing, three variables were retained for analysis, with the first two columns (username and timestamp) excluded due to ethical and legal considerations. The variables included in this dataset are described in Table 2 below.

Table 2.

Description of Variables in the Dataset (Sentimental Analysis)

Variable Name	Description
comment_text	Text of the user comment
username	User's YouTube username (removed)
timestamp	Time when the comment was posted (removed)

This approach enables us to examine both the quantitative shifts in public interest and financial valuation, as well as the qualitative sentiment expressed by consumers, offering a comprehensive understanding of the reputational and market consequences of the scandal.

Data Cleaning

Merging the Datasets

The data was originally presented as individual time series, each containing only a single variable and its corresponding time values. Since direct downloading was not available, the most efficient method for data collection involved copying and pasting the content manually into Excel. The date range filter remained consistent across all searches, which facilitated merging the datasets based on the common time variable (week). This allowed for a straightforward integration of multiple data sources into a single, unified dataset.

Figure 1.

Google Trends Search Overview for Selected Keywords.

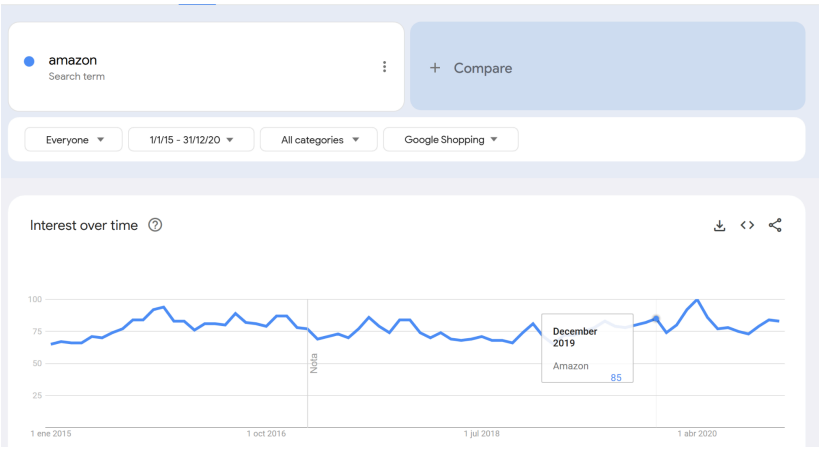


Figure 2.

Consolidated Dataset After Merging All Sources

Week	amazon: (Google Shopping)	amazon: (News search)	tax: (News search)	amazon: (Youtube search)	Open	High	Low	Close	Adj Close	Volume
2015-12-27	84	18	5	28	33.28	34.82	33.28	33.79	33.79	335,744,000
2016-01-03	77	18	7	22	32.81	32.89	30.26	30.35	30.35	661,082,000
2016-01-10	76	18	6	23	30.62	31.3	28.26	28.51	28.51	645,868,000
2016-01-17	75	18	6	23	28.85	30	27.36	29.82	29.82	456,918,000
2016-01-24	75	18	8	22	29.9	31.9	28.5	29.35	29.35	840,150,000
2016-01-31	74	17	9	21	28.91	29.09	24.96	25.11	25.11	772,476,000

Missing values

An inspection was conducted to assess the presence of missing values across all columns. As shown in Figure 2, the analysis confirmed that there were no missing values in the dataset.

Figure 3.

Python code used to detect missing values and the resulting output.

```
# Check for missing values
print("\nMissing values before handling:")
print(df.isnull().sum())

# Forward-fill missing values (for time-series data)
df.fillna(method='ffill', inplace=True)
df.fillna(method='bfill', inplace=True)  # Backward fil

print("\nMissing values after handling:")
print(df.isnull().sum())
```

```
Missing values before handling:
Week                          0
amazon: (Google Shopping)     0
amazon: (News search)         0
tax: (News search)            0
amazon: (Youtube search)      0
Open                          0
High                          0
Low                           0
Close                         0
Adj Close                     0
Volume                        0
dtype: int64
Missing values after handling:
Week                          0
amazon: (Google Shopping)     0
amazon: (News search)         0
tax: (News search)            0
amazon: (Youtube search)      0
Open                          0
High                          0
Low                           0
Close                         0
Adj Close                     0
Volume                        0
dtype: int64
```

Outlier Detection

To detect outliers, it was decided to use the Interquartile Range (IQR) method. First, the code selected the columns with numerical data types using df.select_dtypes. Then, for each numerical column, it calculated the first (Q1) and third (Q3) quartiles to compute the IQR (Q3 - Q1). Based on this, lower and upper bounds were established using the formulas Q1 - 1.5IQR and Q3 + 1.5IQR. Values falling outside these thresholds were identified as outliers. To reduce their impact on the analysis, those below the lower bound and above the upper bound were capped at the respective thresholds using np.where.

Figure 4.

Code to detect Outliers.

```
# Detect and cap outliers using IQR (for numerical columns)
numerical_cols = df.select_dtypes(include=[np.number]).columns

for col in numerical_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    # Cap outliers
    df[col] = np.where(df[col] > upper_bound, upper_bound,
                       np.where(df[col] < lower_bound, lower_bound, df[col]))

    print(f"\nOutliers capped in column: {col}")
```

```
Outliers capped in column: amazon: (Google Shopping)

Outliers capped in column: amazon: (News search)

Outliers capped in column: tax: (News search)

Outliers capped in column: amazon: (Youtube search)

Outliers capped in column: Open

Outliers capped in column: High

Outliers capped in column: Low

Outliers capped in column: Close

Outliers capped in column: Adj Close

Outliers capped in column: Volume
```

Feature Scaling

To standardize the data, a StandardScaler object was initialized. This tool standardizes features by removing the mean and scaling them to unit variance. Then, the numerical features, excluding date-related columns, were transformed using this scaler. This step was performed to normalize the range of the variables, which is essential for improving the performance and stability of many machine learning algorithms.

Figure 5.

Code and summary of feature scaling.

```
# Scale numerical features (excluding date columns)
scaler = StandardScaler()
scaled_values = scaler.fit_transform(df[numerical_cols])
df_scaled = df.copy()
df_scaled[numerical_cols] = scaled_values

print("\nFeature scaling applied to numerical columns.")
```

```
Feature scaling applied to numerical columns.

Explanation:

Standardizes data to mean=0 and variance=1 for machine learning.
```

```
# Extract time-based features
df['Week'] = pd.to_datetime(df['Week'])  # Ensure 'Week' is datetime
df['Year'] = df['Week'].dt.year
df['Month'] = df['Week'].dt.month
df['Week_of_Year'] = df['Week'].dt.isocalendar().week

print("\nAdded time-based features:")
print(df[['Week', 'Year', 'Month', 'Week_of_Year']].head())
```

```
Added time-based features:
        Week  Year  Month  Week_of_Year
0 2015-12-27  2015     12            52
1 2016-01-03  2016      1            53
2 2016-01-10  2016      1             1
3 2016-01-17  2016      1             2
4 2016-01-24  2016      1             3
```

Figure 6.

Dataset before and after cleaning and preparation for machine learning.

Before

ıg)	amazon: (News search)	tax: (News search)	amazon: (Youtube search)	Oper
84	18	5	28	33.2
77	18	7	22	32.8
76	18	6	23	30.6
75	18	6	23	28.8
75	18	8	22	29.9

After

: (News search)	tax: (News search)	amazon: (Youtube search)	Open	High	Low	Close	Volume	Year	Month	Week_
18.0	5.0	28.0	33.28	34.82	33.28	33.79	335,744,000	2015	12	
18.0	7.0	22.0	32.81	32.89	30.26	30.35	661,082,000	2016	1	
18.0	6.0	23.0	30.62	31.30	28.26	28.51	645,868,000	2016	1	
18.0	6.0	23.0	28.85	30.00	27.36	29.82	456,918,000	2016	1	
18.0	8.0	22.0	29.90	31.90	28.50	29.35	840,150,000	2016	1	

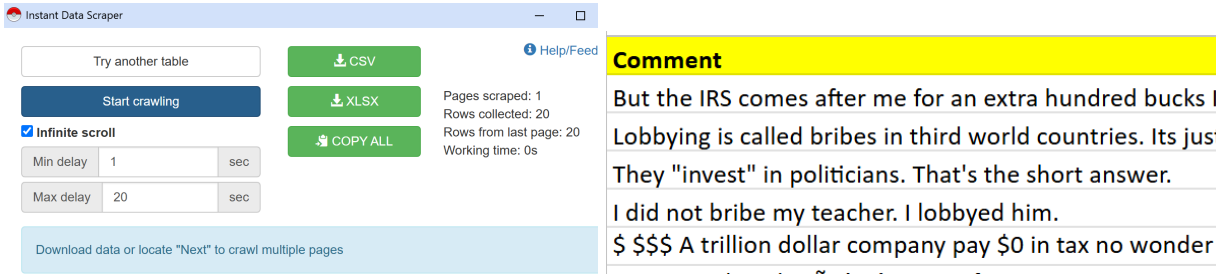
Sentiment Analysis

Data Collection Process

User comments were extracted from a selected YouTube video using Instant Data Scraper. This process took approximately one hour and yielded over 1,500 comments.

Figure 7.

Code used for comment extraction and a preview of the raw dataset.

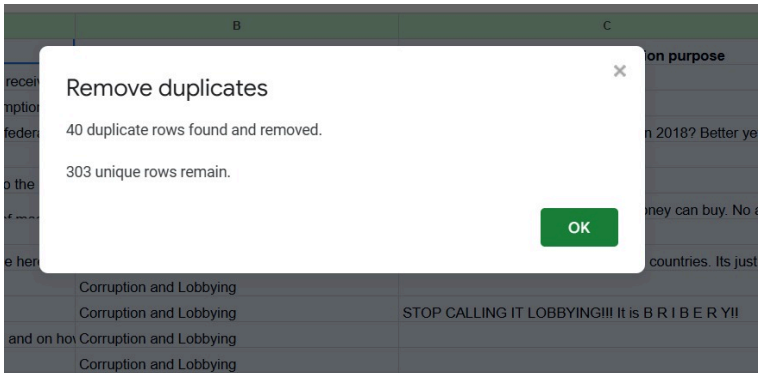


Data Cleaning

Automated duplication errors generated by the scraping tool were removed using a custom function in Google Sheets.

Figure 8.

Screenshot showing the number of duplicate comments removed.



Next, the text data was preprocessed using **Python**. This included converting all text to lowercase and removing non-alphabetic characters to ensure uniformity and consistency for downstream analysis.

Figure 9.

Python code used for text normalization.

```
# Removing hashtags and mentions
def get_hashtags(text):
    hashtags = re.findall(r'\#\w+',text.lower())
    return hashtags
def get_mentions(text):
    mentions = re.findall(r'\@\w+',text.lower())
    return mentions

# Cleaning up the text of the tweets
def remove_content(text):
    text = re.sub(r"http\S+", "", text) #remove urls
    text=re.sub(r'\S+\.com\S+','',text) #remove urls
    text=re.sub(r'\@\w+','',text) #remove mentions
    text =re.sub(r'\#\w+','',text) #remove hashtags
    return text

def process_tweet(tweet):
    """
    tweets cleaning by
    1) lowering the case of the tweet,
    2) removing unwanted symbols and replacing them with a whitespace,
    3) split sentences into words according to whitespaces and then
    4) join back with a single whitespace as separator between various words
    """
    return " ".join(re.sub("(@[A-Za-z0-9]+)|([^0-9A-Za-z \t])", " ",tweet.lower()).split())

def process_text(text, stem=False): #clean text
    text=remove_content(text)
    lemmatizer=WordNetLemmatizer()
    text = re.sub('[^A-Za-z]', ' ', text.lower()) #remove non-alphabets
    text = re.sub(r'@[A-Za-z0-9]+', '', str(text)) # remove @mentions
    text = re.sub(r'#', '',  str(text)) # remove the '#' symbol
    text = re.sub(r'RT[\s]+', '',  str(text)) # remove RT
    text = re.sub(r'https?\/\/S+', '',  str(text)) # remove the hyperlink
    text = re.sub(r'http\S+', '',  str(text)) # remove the hyperlink
    text = re.sub(r'www\S+', '',  str(text)) # remove the www
    text = re.sub(r'pic+', '',  str(text)) # remove the pic
    text = re.sub(r'com', '',  str(text)) # remove the pic
    text = re.sub(r"\bamp\b", ' ', text.lower()) #remove "amp" which is coming from the translation of &
    text = re.sub(r"\bco\b", ' ', text.lower()) #remove "co" which was one of the top words found below
    tokenized_text = word_tokenize(text) #tokenize
    #tokenized_text = [lemmatizer.lemmatize(word) for word in tokenized_text]
    clean_text = [
         word for word in tokenized_text
         if (word not in stop_words and len(word)>1)
```

Figure 10.

Comment text before and after preprocessing.

Before	*After*
Comment	**cleaned_comments**
All companies receive incentives. CNBC's paren...	all companies receive incentives cnbc s parent...
Congress writes tax law. They will never refor...	congress writes tax law they will never reform...
How much federal tax did Trump pay in 2018? Be...	how much federal tax did trump pay in 2018 bet...
Is called corruption sweetie, look it up	is called corruption sweetie look it up
It's telling that Amazon placed it's new HQ in...	it s telling that amazon placed it s new hq in...

Topic Classification

A manual active learning approach was employed for comment classification. Using the cleaned dataset, the team first prompted ChatGPT to generate an initial set of 15 candidate categories. These were then reviewed collaboratively to ensure mutual exclusivity, relevance, and theoretical coherence. After refinement, the final classification scheme included the following categories:

- Criticism of the Tax System
- Corruption and Lobbying
- Moral or Ethical Outrage
- Justification of the Corporate Model
- Reform Proposals or Citizen Action
- Sarcasm or Critical Humor
- International Comparison
- General Institutional Distrust

To operationalize the classification process, a reference script was created containing representative examples for each category. This script was used to configure the ChatGPT interface for consistent annotation across all team members. Then, each member manually selected 20 comments at a time and used ChatGPT to classify them. Results were returned in tabular format. Using this iterative method, each participant classified approximately 150 comments.

Figure 11.

Final classified dataset with comment labels.

Topic	cleaned_comments
Corruption and Lobbying	all companies receive incentives cnbc s parent...
Corruption and Lobbying	congress writes tax law they will never reform...
Corruption and Lobbying	how much federal tax did trump pay in 2018 bet...
Corruption and Lobbying	is called corruption sweetie look it up
Corruption and Lobbying	it s telling that amazon placed it s new hq in...

Polarity and Subjectivity Classification

To assess the emotional tone and subjectivity of user comments, sentiment analysis was conducted using the TextBlob library in Python. Two custom functions were defined to extract polarity and subjectivity scores for each comment. Polarity ranges from –1 (negative) to +1 (positive), while subjectivity ranges from 0 (objective) to 1 (subjective). Based on the polarity score, comments were categorized into three sentiment classes: Negative, Neutral, or Positive. Similarly, subjectivity scores were interpreted to assess the degree of personal opinion versus factual content. An additional variable was created by concatenating both analyses (Analysis_Polarity and Analysis_Subjectivity) into a combined descriptor (Sentiment) for each comment. This classification enabled a structured overview of the emotional and cognitive tone expressed across the dataset.

Figure 12.

Python code used for Polarity and Subjectivity Classification.

```
def getSubjectivity(text):
    return TextBlob( str(text)).sentiment.subjectivity

def getPolarity(text):
    return TextBlob( str(text)).sentiment.polarity

def get_Polarity_Analysis(score):
    if score < 0:
      return 'Negative'
    elif score == 0:
      return 'Neutral'
    else:
      return 'Positive'
def get_Subjectivity_Analysis(score):
    if score >  0:
      return 'Opinion'
    else:
      return 'Fact'

df['Sentiment'] = df.apply(lambda x: ' '.join([str(x['Analysis_Polarity']),str(x['Analysis_Subjectivity'])]),axis=1)
```

Label Encoding for Categorical Variable

The code taught in class was used to perform label encoding for the categorical variable (Topic).

Figure 12.

Code for label encoding of categorical variable.

```
from sklearn.preprocessing import LabelEncoder

label_encoder = LabelEncoder()
df['Topic'] = label_encoder.fit_transform(df['Topic'])
```

Figure 13.

Dataset before and after cleaning and preparation for machine learning Sentiment Analysis.

Before

Comment
But the IRS comes after me for a
Lobbying is called bribes in third
They "invest" in politicians. That'
I did not bribe my teacher. I lobb
$ $$$ A trillion dollar company p

After

Topic	cleaned_comments	Subjectivity	Polarity	Analysis_Polarity	Analysis_Subjectivity	Sentiment
Corruption and Lobbying	all companies receive incentives cnbc s parent...	1.000000	0.600000	Positive	Opinion	Positive Opinion
Corruption and Lobbying	congress writes tax law they will never reform...	0.312500	0.237500	Positive	Opinion	Positive Opinion
Corruption and Lobbying	how much federal tax did trump pay in 2018 bet...	0.300000	0.300000	Positive	Opinion	Positive Opinion
Corruption and Lobbying	is called corruption sweetie look it up	0.000000	0.000000	Neutral	Fact	Neutral Fact

Visualizations, Analytical Methods & Implementation

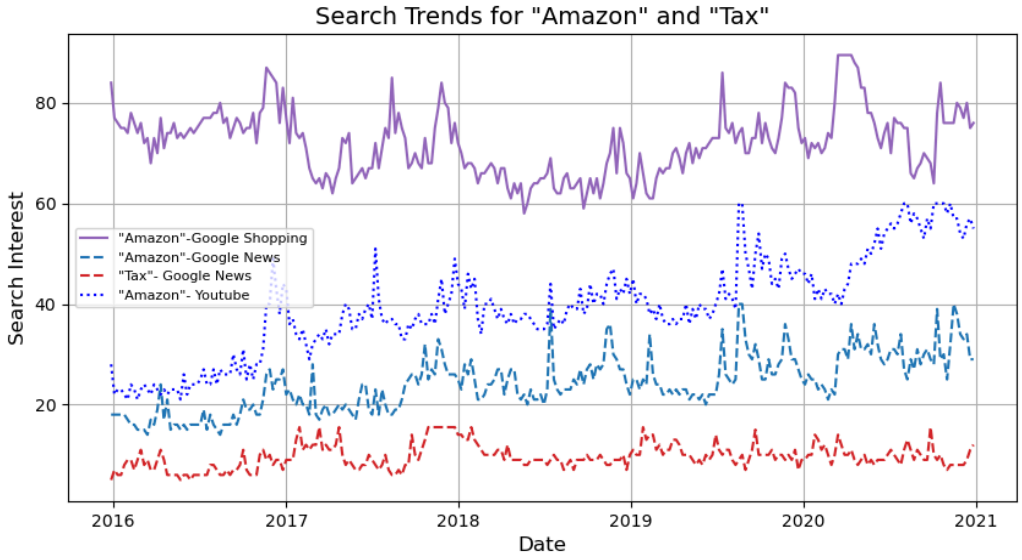
Exploratory Data Analysis

1. ***What were the temporal trends in online searches related to Amazon and taxes, as well as Amazon's stock prices?***

   1.1 What were the search patterns on Google related to Amazon and taxes?

To understand search patterns related to Amazon and taxes, a line chart was chosen because it effectively visualizes how interest in specific search terms changes over time. This type of visualization allows for clear observation of trends, spikes, and dips in public attention, which is crucial for identifying moments of heightened interest or controversy. By plotting search interest for Amazon in Google Shopping searches, Google News searches, Youtube searches, as well as for "Tax" in News searches, the chart provides a comparative view of how public and media engagement evolved from 2016 to 2021. This helps contextualize shifts in awareness and concern surrounding the Amazon tax scandal and supports a deeper understanding of how external events may influence consumer sentiment and online behavior.

Figure 14.
Search Trends for Amazon and Tax



Note: Google Trends values are normalized from 0 to 100, where 100 represents the peak popularity of a search term during the selected time frame. The values shown do not represent absolute search volumes.

Based on the figure titled *Search Trends for "Amazon" and "Tax"*, there is no clear inflection point or sudden spike in Google News searches around 2018 or early 2019, the period when Amazon's tax practices were publicly scrutinized. The search interest for the term “tax” in the News category remains relatively stable throughout the five year window, with no notable changes that could be directly linked to the scandal. And the search for “Amazon” in google shopping had a positive increase through the first semester of 2019.

Seeing the full picture, "Amazon" in Google News exhibits a gradual upward trend, increasing from a normalized value of around 20 in 2016 to just over 30 by 2020. However, this growth appears steady and not concentrated around the scandal’s exposure in early 2019, suggesting that public attention toward Amazon in the news was not drastically altered by the event.

The Google Shopping trend for "Amazon" shows a temporary decline in 2018 compared to 2017, which might initially seem like a reputational impact. However, given the scandal only became widely known in early 2019, and tax disclosures often reference the prior fiscal year, this is likely unrelated. In fact, the Shopping trend recovers sharply in 2019 and continues to grow, indicating that consumer demand remained strong and unaffected by ethical concerns.

The YouTube search trend for "Amazon" also reveals a gradual increase over the five year period, with a noticeable upward shift beginning in late 2019 and continuing through 2020. This may reflect the rising volume of Amazon related content, such as product reviews and

unboxings, particularly during the pandemic when digital content consumption increased. However, like the News search trends, this increase does not align directly with the timeline of the tax scandal and appears to follow broader behavioral trends.

Importantly, Google Shopping has consistently been the most popular platform for "Amazon" related searches, significantly surpassing News and YouTube in relative search interest. In conclusion, this figure may illustrate a broader pattern in public behavior: while ethical scandals can spark temporary awareness, consumer memory and behavior tend to normalize quickly, especially when core product demand remains unaffected. For stakeholders in Amazon, the continued growth in Google Shopping searches may ultimately carry more weight than momentary reputational risks in the News category.

1.2 How did Amazon's stock price evolve before, during, and after the scandal?

A line chart was also chosen to answer this question because it provides a clear, continuous view of change over time. In this case, it effectively illustrates the trend in Amazon's stock price, making it easy to observe growth patterns, fluctuations, and notable shifts that may align with public sentiment or media coverage. Its time-based layout complements the search trend chart, allowing for intuitive comparison across both public interest and financial performance.

Figure 15.
Amazon Stock Price Over Time



As shown in the figure *"Amazon Stock Price Over Time"*, Amazon's stock experienced a strong upward trend from 2016 to 2020, rising from approximately **$30** in early 2016 to over **$160** by the end of 2020. This reflects significant long-term growth and investor confidence.

During the second half of 2018, however, the stock price fell from around **$100** to **$80**, representing a decline of approximately 20%. While this drop coincides with the period leading up to Amazon's tax avoidance controversy (publicly reported in February 2019), it is more likely attributable to broader macroeconomic and political factors. These include:

- Heightened market volatility due to the U.S. and China trade war initiated by the Trump administration,
- Falling oil prices in the latter half of 2018,
- Concerns over the Federal Reserve's interest rate hikes, which triggered a wider stock market correction across tech stocks.

Given these overlapping events and the timeline of the scandal, we do not attribute the decline of the stock prices to the tax avoidance scandal.

Then in the plot it can be seen that Amazon's stock price recovered and stabilized throughout 2019, indicating that investor confidence remained intact. Notably, in 2020, the company experienced a substantial surge in stock value, closely tied to the COVID-19 pandemic. With lockdowns in effect and public concern over virus exposure, e-commerce demand spiked, driving Amazon's revenues and stock price to new highs.

In conclusion, the tax avoidance scandal did not result in a lasting negative impact on Amazon's stock performance. While public interest and media coverage may have increased, the market appears to have interpreted Amazon's tax strategy as legally permissible and financially

advantageous. The overall trajectory of the stock suggests that economic fundamentals and strategic positioning outweighed reputational concerns in the eyes of investors.

Statistical Data Analysis

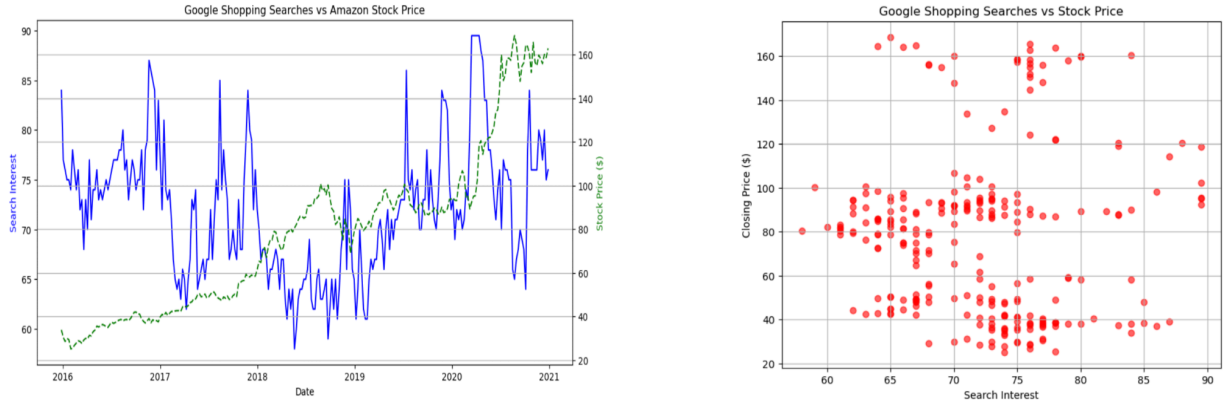
2. ***Is there a correlation between online interest and Amazon's commercial value?***

By plotting Google Shopping search interest against Amazon's closing price, it's easier to visually determine any correlation or clustering. The scatter plot doesn't account for time, allowing for a focused view on the relationship between the two variables.

Both plots provide complementary perspectives: the line plot shows temporal trends, while the scatter plot focuses on direct relationships between the variables. Together, they give a comprehensive view of the potential correlation between search interest and stock price.

Figure 16.

Dual-axis plot and scatter plot of shopping searches vs stock price



The comparison between Google Shopping search interest and Amazon's stock price reveals an inconsistent relationship. While Amazon's stock price shows a clear upward trajectory over time, the search interest is much more volatile, with noticeable fluctuations and no sustained upward trend. The scatter plot further supports this, displaying a weak and dispersed relationship

between the two variables. This suggests that consumer search activity on Google Shopping does not reliably reflect or predict Amazon's commercial performance in the stock market. Although both may be influenced by broader trends, online interest alone does not appear to be a strong indicator of investor value.

Figure 17.

Heatmap of correlations between search trends and stock price



The correlation matrix shows that YouTube search interest for "Amazon" has a strong positive correlation with Amazon's stock price (0.85), suggesting a meaningful association between public engagement with Amazon-related video content and the company's market value. News search interest also shows a moderate correlation (0.69), indicating that increased media attention may align with periods of stock growth.

On the other hand, Google Shopping search interest has virtually no correlation (-0.003) with stock price, implying that consumer browsing behavior does not significantly reflect or influence investor sentiment. Overall, engagement through platforms that reflect public attention (like YouTube and news) appears to be more connected to Amazon's commercial performance than transactional search behavior.

Machine Learning Analysis

3. *What extent do search engine trends influence the commercial value of Amazon?*

Technique: Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was selected as a dimensionality reduction technique to examine whether the number of features in the dataset could be reduced while still preserving most of the variance. PCA works by transforming the original features into new, uncorrelated components (called principal components) that capture the most variance in the data. This technique is especially useful when dealing with high-dimensional data, as it can improve model performance and training efficiency while helping reveal hidden structures.

Figure 18.

PCA implementation code and a preview of the resulting DataFrame with the principal components.

```
# One Hot Encoding

df = pd.get_dummies(df, columns = ['Month'], drop_first=True)
```

```
# Select features and target variable
X = df.drop(columns=['Open', 'High', 'Low', 'Close'])
y = df['Close']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Standardize the features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
# Convert the scaled data back to a DataFrame for easier inspection
X_train_scaled_df = pd.DataFrame(X_train_scaled, columns=X_train.columns)

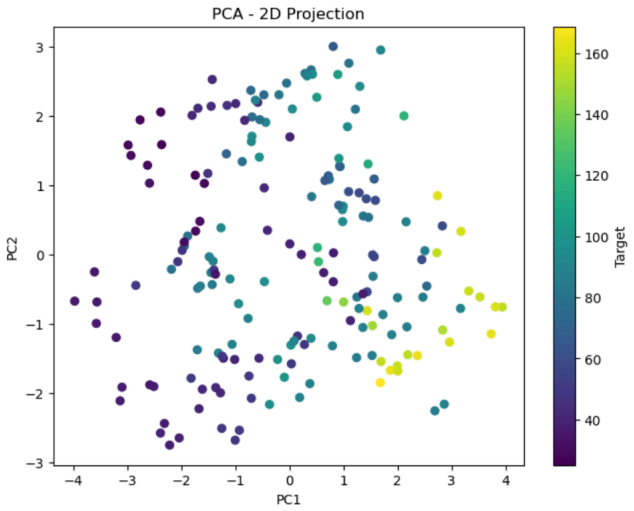
# Display the first 3 rows of the standardized training data
print("First 3 rows of standardized training data:")
print(X_train_scaled_df.head(3))
```

```
First 5 rows of PCA results:
        PC1       PC2
0 -1.170506  1.457850
1 -0.221573 -1.510977
2 -1.156529  2.157228
3 -1.740772  0.343299
4 -1.949014  0.129083
```

The summary of the PCA transformation shows how the data is redistributed across the principal components. However, this alone does not allow us to determine whether PCA was beneficial for this specific dataset.

Figure 19.

2D PCA plot and cumulative variance summary



	Explained Variance	Cumulative Variance
PC1	0.163350	0.163350
PC2	0.117543	0.280893

The PCA visualization and variance table suggest that while the first two principal components (PC1 and PC2) capture only about 28% of the total variance, there is still some observable structure in the data. The color gradient representing Amazon's stock price reveals clusters of higher values, particularly toward the right side of the plot, indicating that certain combinations of search trend features may align with stock performance. However, despite this visual indication of possible patterns, the explained variance remains relatively low, with PC1 accounting for 16.34% and PC2 for 11.75% of the variance.

This means that the first two components together explain just 28.09% of the dataset's variability. A cumulative variance this low suggests that capturing a meaningful amount of total information (e.g., 90–95%) would require many more principal components, which defeats the purpose of dimensionality reduction. As a result, while PCA was useful for exploring the data

and generating preliminary insights, it was not used for model training, as the transformation does not meet the minimum requirements for preserving sufficient variance.

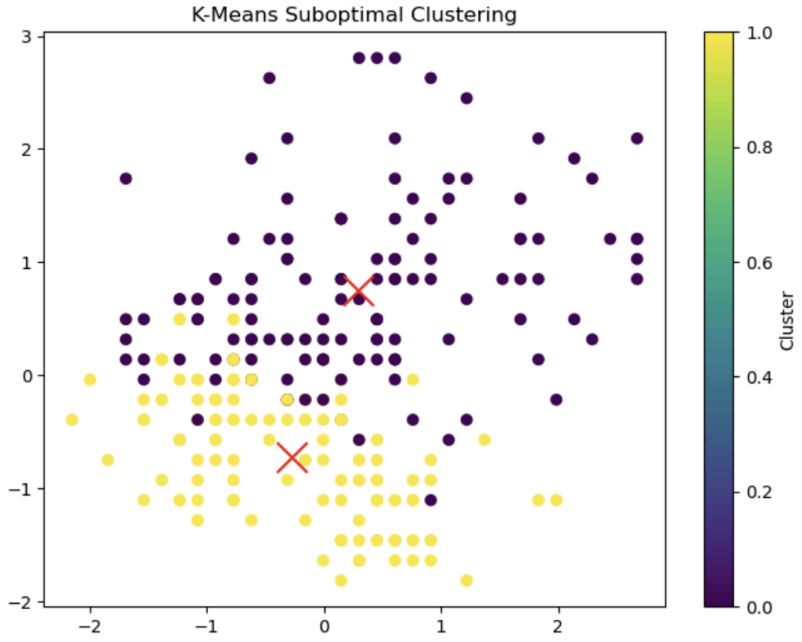
Technique: K-Means Clustering

To detect potential segments, we applied K-Means clustering, an unsupervised algorithm that segments data into distinct groups (clusters) by minimizing the distance between points and their assigned centroids. This approach was chosen for its simplicity and scalability in grouping similar observations.

Before performing clustering, the dataset underwent preprocessing: categorical variables were encoded using OneHotEncoder to convert them into a numerical format suitable for K-Means, and numerical features were standardized using StandardScaler to ensure equal weighting. The clustering process was evaluated using both the Silhouette Score, which measures how similar an object is to its own cluster compared to other clusters, and Inertia, which measures within-cluster sum-of-squares.

Figure 20.

K-Means Suboptimal Clustering



The K-means clustering visualization with two clusters reveals a visible separation between data points, suggesting that search engine trends can be grouped into distinct behavioral patterns that may relate to fluctuations in Amazon's stock price. While the clustering is not perfectly clean, the formation of two dominant groups implies some underlying structure in the data. This supports the idea that public interest or sentiment reflected in search activity could influence or coincide with shifts in commercial performance. Although the clusters may not represent precise market predictions, the emergence of distinct groupings reinforces the potential of machine learning to uncover nonlinear relationships that traditional analysis might overlook.

Technique: Regression Model Building

Regression was also explored to predict Amazon's closing stock price based on features derived from search engine trends. Linear Regression was included as a baseline due to its simplicity, interpretability, and effectiveness in modeling linear relationships. In addition, other regression models were evaluated (KNN, Support Vector Regression, and XGBoost) to compare performance across algorithms with varying complexity and ability to capture non-linear patterns.

Linear Regression estimates the relationship between input features and a continuous target by fitting a line that minimizes squared error. KNN predicts outcomes based on the average of the nearest neighbors in feature space, making it useful for non-linear patterns. Support Vector Regression (SVR) seeks to find a function that fits the data within a specified margin of tolerance, focusing on maximizing the margin around the predicted line and minimizing errors outside that margin. XGBoost uses an ensemble of decision trees built sequentially to correct errors from previous iterations, capturing complex relationships and

interactions in the data. This range of models enables a comprehensive evaluation of whether and how search trends can effectively predict stock performance.

Figure 21.

Model Performance Comparison (R^2 Scores)

```
LR: 0.8976
KNN: 0.6456
SVR: 0.2144
XGB: 0.9856
```

Although XGBoost achieved the highest R^2 score of 0.9856, we selected Linear Regression as the preferred model for further analysis due to its simplicity, interpretability, and transparency. While XGBoost is highly accurate, its complexity makes it harder to interpret and communicate results to non-technical audiences. In contrast, Linear Regression performed well with an R^2 of 0.8976 and offers clear insights into how input features influence predictions, which is especially valuable when explaining how search trends may relate to stock performance.

A scatter plot of actual vs. predicted closing prices using Linear Regression was then used to visually evaluate the Linear Regression model's performance. This plot provides a direct comparison between the true stock values and those predicted by the model, helping us understand how closely the predictions align with reality. The red dashed line representing perfect prediction serves as a visual benchmark, making it easy to assess whether the model is systematically under- or over-predicting.

Figure 22.

Linear Regression: Actual vs Predicted Closing Price



The visualization shows that most points are tightly clustered around the red line, indicating that the Linear Regression model predicts Amazon's stock price with a high degree of precision and alignment with actual values. While a few deviations exist, especially at higher price values, the general alignment with the perfect prediction line reinforces that search engine trends contain meaningful signals related to stock performance. This supports the use of Linear Regression as a reliable and interpretable tool for linking public interest to commercial outcomes.

Sentiment Analysis

This study analyzes public perception towards Amazon based on the comments generated by a video related to its 2018 tax controversy. Although not all comments mention the company explicitly, it is assumed that, given the communicative and thematic context of the video, all the identified reaction categories are directly or indirectly related to Amazon. For example,

criticisms of the tax system, corruption or lack of corporate ethics, while they might seem structural, are interpreted as a form of rejection of the actors who benefit from these dynamics, among which Amazon is the implicit protagonist.

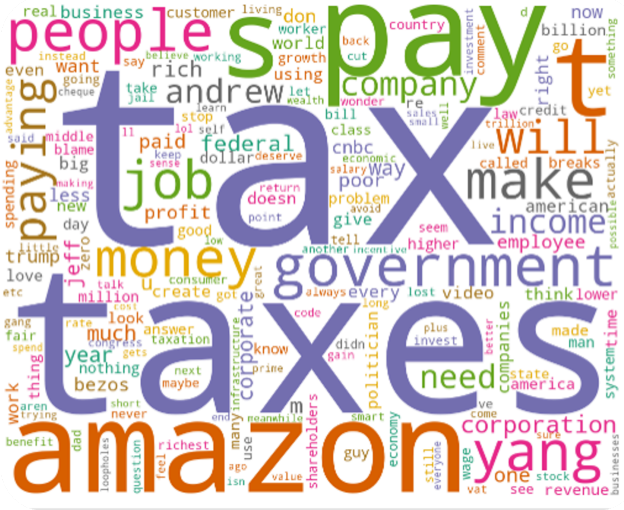
Therefore, in this analysis, all negative reactions are interpreted as indicators of negative perception towards the brand, and all positive ones as expressions of justification or defense of Amazon, thus allowing to address the objective of studying the association between sentiment polarity and brand perception.

4. ***How do the sentiment polarity and thematic patterns in viewer comments reflect public opinion regarding Amazon's tax scandal?***

To identify the primary topics associated with Amazon's tax scandal, a word cloud visualization was used. This method was chosen because it offers a clear and immediate view of the most frequently mentioned terms within a large body of text data such as user comments, articles, or social media posts by scaling words based on frequency. The visual simplicity of a word cloud makes it especially useful for exploratory analysis, allowing us to quickly identify dominant themes without needing to read through thousands of individual entries.

Figure 23.

Word frequency plot of primary topics discussed in YouTube comments



The word cloud visualization reveals several key findings about public discourse surrounding Amazon's tax scandal. The most prominent terms are "tax," "taxes," and "pay," indicating a high frequency of tax-related concerns. The central placement of "Amazon" highlights that the conversation was strongly focused on the company itself and its perceived role in the issue. Additionally, words such as "people," "job," "money," and "corporation" suggest that many users were comparing the experiences of ordinary taxpayers with the advantages enjoyed by large corporations, reflecting broader concerns about fairness. The appearance of terms like "government," "federal," and "yang" shows that the conversation extended into political critique, pointing to dissatisfaction not just with Amazon but with the broader system that enables such practices. Finally, the presence of words like "need," "make," "change," and "breaks" signals a desire for reform, indicating that many commenters were not only critical but also calling for changes in tax policy or corporate accountability.

To evaluate the overall sentiment polarity and subjectivity of the dataset and assess the general tone of public opinion, a combination of visualizations was used. The scatter plot was chosen to provide a continuous view of how individual comments are distributed across the spectrum of polarity and subjectivity scores, helping to uncover patterns and relationships in emotional tone and factual content. In addition, the bar charts were used to give a clear, aggregated breakdown of sentiment polarity and subjectivity categories, allowing for quick quantification and comparison. This combination offers both detailed insight into individual sentiment behavior and a broad summary of the general sentiment distribution, making the analysis more comprehensive and visually intuitive.

Figure 24.

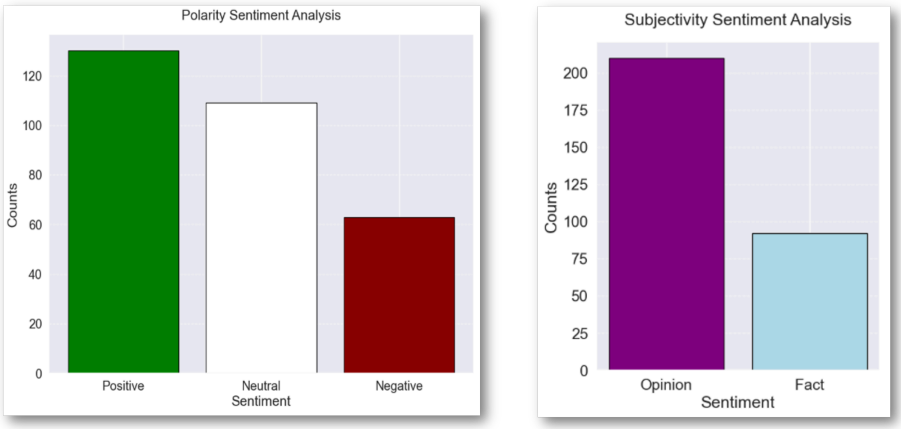
Scatter plot of polarity vs. subjectivity



The scatter plot shows a clustering of comments around the neutral to positive polarity region, with many comments appearing in the center-right quadrant. This suggests that while some users expressed negative sentiment, the overall tone leans slightly positive. The vertical spread in subjectivity reveals a balanced mix of emotionally driven and more fact based comments, with a slight tendency toward subjective content.

Figure 25.

Bar charts of polarity and subjectivity distributions

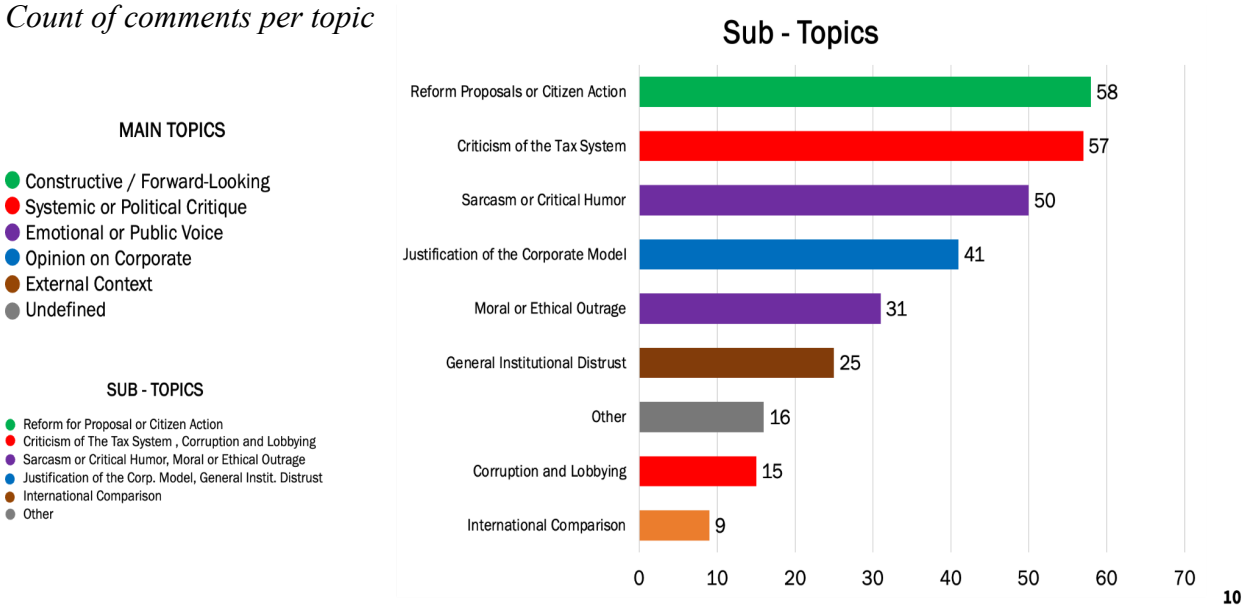


The polarity bar chart shows that positive sentiment has the highest count, followed by neutral sentiment, with negative sentiment being the least represented. This indicates that public opinion, although mixed, tends to lean more favorably than critically toward the topic or brand in question. The subjectivity bar chart shows that opinion based comments are far more frequent than factual ones. This reflects a conversational and emotionally expressive tone in the public discourse, suggesting that users are sharing more personal reactions and viewpoints than objective or verifiable claims.

To explore which categories of viewer comments generated the highest engagement based on the volume of responses per topic, a horizontal bar chart was selected. This visualization was chosen because it clearly displays and compares the frequency of responses across multiple subtopics, allowing for easy identification of the most and least engaging themes. The horizontal layout also ensures that lengthy topic labels remain readable, while the use of distinct colors improves visual differentiation between categories. This format is especially effective when working with categorical data, as it emphasizes relative magnitudes and makes patterns in user engagement more immediately apparent.

Figure 26.

Count of comments per topic



The bar chart reveals that the most engaging comment categories were Reform Proposals or Citizen Action and Criticism of the Tax System, each drawing over 55 responses. These high counts suggest that viewers are highly responsive to calls for change and critiques of the current system. Sarcasm or Critical Humor also ranked highly, indicating that users often engage with content that mixes humor with criticism. On the other hand, topics like International Comparison and Corruption and Lobbying received fewer responses, suggesting they were less central to public discussion. Overall, the chart highlights that audiences are especially engaged by discussions that challenge the status quo or propose alternatives, reinforcing the importance of understanding which narratives resonate most with the public.

5. ***To what extent do specific categories of comments and their interconnections influence consumer attitudes and perceptions toward Amazon's brand?***

To determine how specific categories of comments contribute to shaping consumer attitudes toward Amazon, Polarity and Subjectivity scatterplots by categories are presented. By plotting polarity scores, which moves from negative to positive sentiment, against subjectivity levels, we are capturing the emotional tone and the subjectivity intensity that online users express within each specific topic. The color gradient (from red to green) provides an immediate visual guide of sentiment direction, where negative is red, neutral is white, or positive is green, while the position on the y axis (subjectivity) shows how much the content is driven by personal opinion rather than factual reporting. By comparing emotional tone across themes, we can identify outliers or patterns, and better understand how different types of discourse contribute to public perceptions of Amazon.

Figure 27.

Top three categories with negative sentiment score

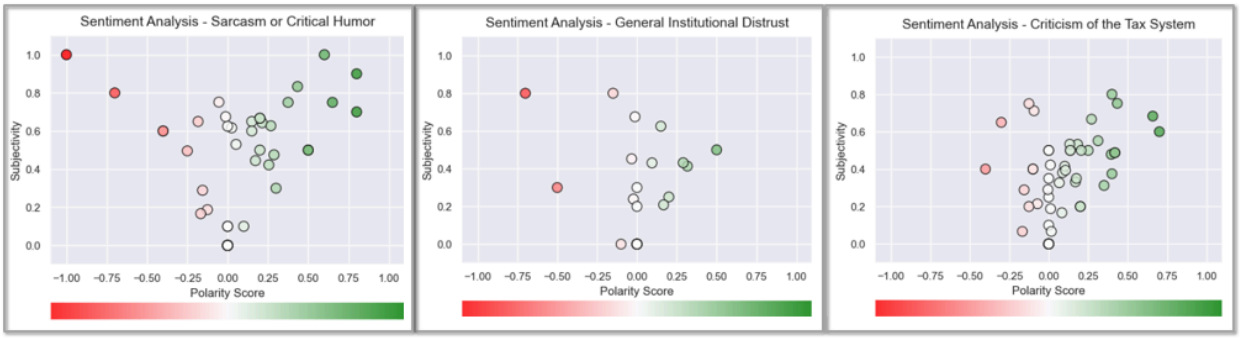


Figure 28.

Categories with Neutral or Balanced Sentiment

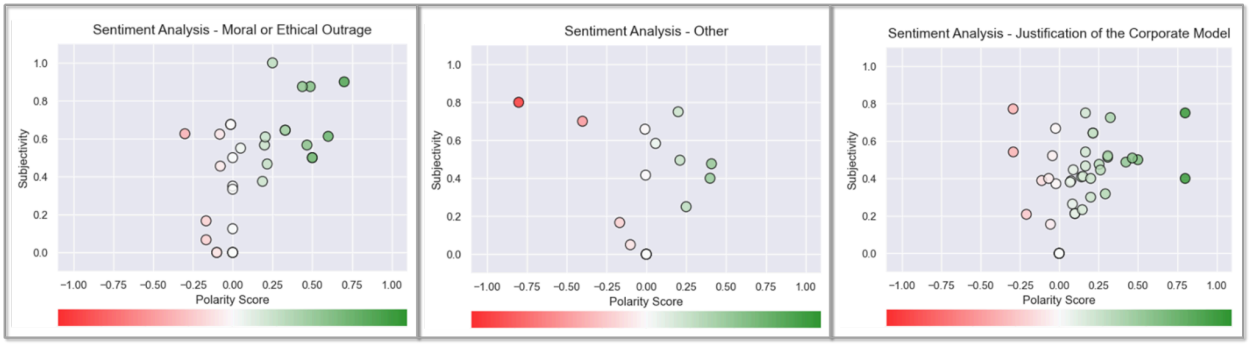
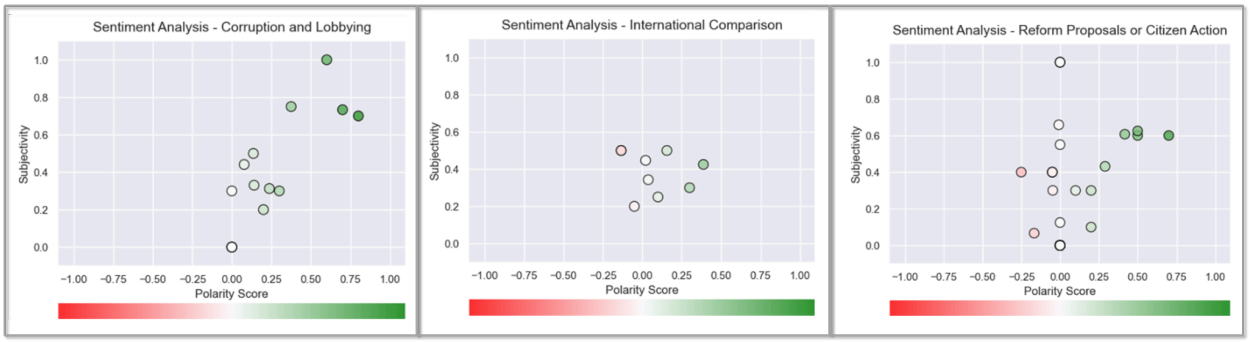


Figure 29.

Categories Dominated by Positive Sentiment



To address the objective of determining how specific categories of comments contribute to shaping consumer attitudes toward Amazon, the sentiment analysis results were organized into three thematic groups based on the polarity distribution within each category.

1. Categories Dominated by Negative Sentiment

This first group includes the categories Sarcasm or Critical Humor, General Institutional Distrust, and Criticism of the Tax System, topics that displayed the highest frequency of negative sentiment values. However, it is important to note that being part of this group does not imply domination by negative sentiment overall. For instance, in Sarcasm or Critical Humor, most comments actually cluster around neutral or even slightly positive scores. This happens because the sentiment model struggles with sarcasm. Since it relies mainly on detecting strong negative words, it often misses sarcastic comments that sound neutral on the surface but are actually very critical. All three categories in this group also show high subjectivity, meaning they're mostly based on personal opinions rather than facts or data.

2. Categories with Neutral or Balanced Sentiment

The second group includes Moral or Ethical Outrage, Justification of the Corporate Model, and Other. These topics have a more balanced mix of positive and negative sentiment, or tend to cluster around neutral. An interesting case is Moral or Ethical Outrage, even though it sounds like a category that would carry negative emotion, the model often classifies these comments as more positive. This may be because people express their concerns using calm or emotionally neutral language, which can make the sentiment harder to detect. This shows how the tone of voice can affect how sentiment is measured.

3. **Categories Dominated by Positive Sentiment**

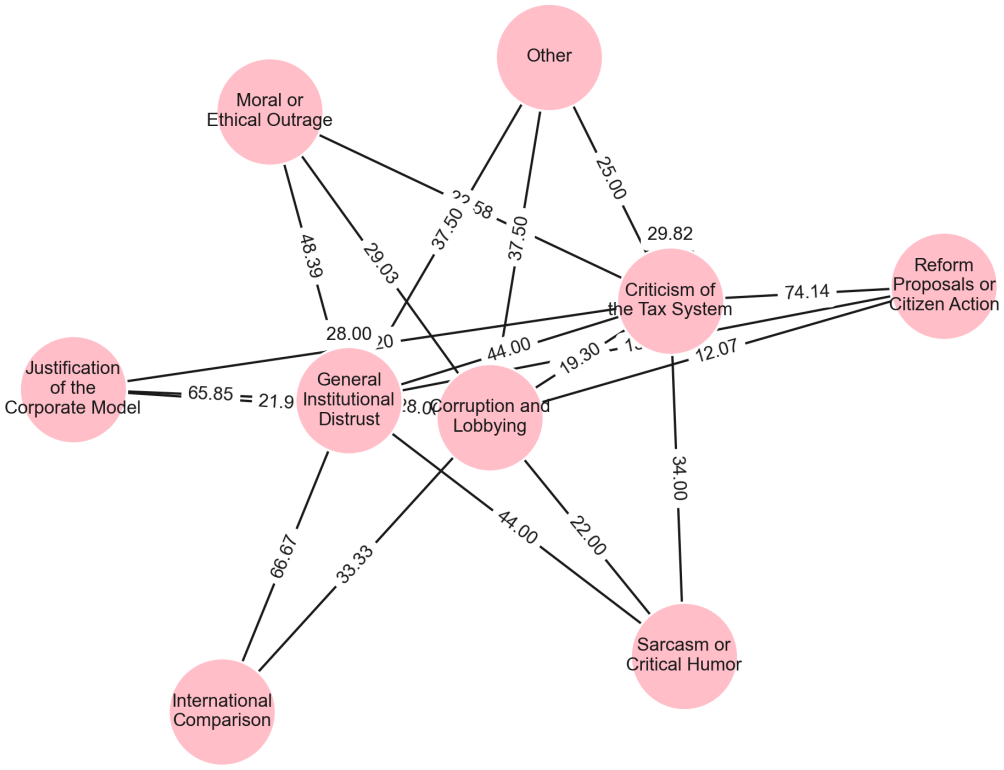
In the last group, Corruption and Lobbying, International Comparison, and Reform Proposals or Citizen Action, most of the sentiment scores lean toward the positive side. While Reform Proposals or Citizen Action being positive makes sense, since people are sharing ideas for change or expressing hope, it's surprising to see Corruption and Lobbying scored that way. This is likely because words like "corruption" don't get flagged as negative by the model unless they're paired with clearly negative words like "angry" or "terrible." So, a comment like "Bezos is corrupt" might not be picked up as strongly negative. It's a reminder that models don't always pick up the deeper meaning if it isn't expressed with emotional words.

Across all categories, the majority of comments are subjective rather than factual, which is expected in social media viral videos. However, categories such as Criticism of the Tax System, Justification of the Corporate Model, and International Comparison exhibit relatively lower subjectivity. This suggests that users in these categories may be drawing from technical knowledge or structural arguments, often involving policy, economics, or comparative insights, which require a more objective and informed tone.

To map the structure and interconnections among key discussion topics to uncover the dynamics behind the audience sentiment, network plot was generated, because it capture how different thematic categories, such as "Criticism of the Tax System," "Corruption and Lobbying," and "General Institutional Distrust", are interconnected through shared co-occurrence in public discourse. The nodes represent distinct discussion topics, while the edges indicate the strength or frequency of their co-occurrence, making visible how often two themes appear together in the same context.

Figure 30.

Degree of Centrality and Betweenness of Youtube Topic Comments



The network visualization reveals key structural insights into how discussion topics around Amazon's tax practices are interconnected. Criticism of the Tax System emerges as the most central node in the network, connecting with the highest number of other themes. This indicates that it functions as a thematic bridge, frequently appearing in conversations that span multiple angles of the scandal.

Corruption and Lobbying also stands out with high betweenness, linking more fragmented or polarized topics such as Moral or Ethical Outrage, General Institutional Distrust, and Other. This suggests it plays a critical role in maintaining the flow between different tones of discourse, whether factual, emotional, or satirical.

In contrast, topics like International Comparison and Justification of the Corporate Model appear on the periphery of the network, implying they are more specific, contributing to isolated rather than central threads of discussion.

Overall, the network structure highlights how certain themes, particularly Criticism of the Tax System and Corruption and Lobbying, anchor public sentiment, acting as hubs where diverse viewpoints converge. These findings make evident the focal points of public concern and the narratives that shape collective reactions to Amazon's tax behavior.

Insights And Discussion

The analysis of online search interest and public discourse surrounding Amazon revealed key insights into how digital attention correlates with brand perception and potential reputational risks. Notably, YouTube search interest for "Amazon" exhibited a strong positive correlation ($r = 0.85$) with the company's stock price, higher than the correlations observed for Google News or Google Shopping. This suggests that user engagement on YouTube may serve as a valuable early indicator of brand value and public interest, underscoring the strategic importance of monitoring this platform.

In contrast, Google News searches demonstrated a negative association with stock performance, indicating that increased news activity may reflect or even amplify reputational risks. This divergence between platforms suggests that while YouTube often reflects consumer engagement and interest, Google News may serve as a proxy for media scrutiny and emerging crises. Therefore, Amazon and similar brands should prioritize continuous monitoring of news cycles, particularly during periods of heightened attention, as negative headlines can rapidly influence stakeholder perceptions and market outcomes.

Network analysis of online discourse offered deeper insight into the structure and drivers of reputational risk. Rather than "Amazon" itself being the central focus of negative conversations, terms like "Tax System Criticism" emerged as key nodes. These topics were densely connected to discussions of corruption and fairness, indicating that public backlash was not isolated to the brand but was embedded in broader socio-political narratives. This framing transforms isolated brand criticism into a reputational hotspot within the network, making early detection and rapid response strategies critical for damage control.

Despite the negative discourse, the data also revealed a subset of users consistently defending Amazon in online conversations. These advocacy voices, although less numerous, offer an opportunity for reputation recovery. By identifying and strategically amplifying these loyal customer narratives, brands can offset part of the negative sentiment and reestablish trust with broader audiences.

Limitations of this analysis include the use of normalized Google Trends data, which limited access to absolute search volumes and reduced the granularity of user behavior insights. Additionally, sentiment analysis tools had difficulty interpreting sarcasm and subtle criticism, potentially leading to an underestimation of negative sentiment. The YouTube comment analysis was also limited, as it focused on a single video related to a specific scandal, which may not reflect broader public opinion. To address these limitations, future research should examine multiple videos, incorporate analysis of news headlines, and track social media content across platforms to capture the full scope and framing of public reactions.

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